

# Siddharth Vijaymurugan

Durham, NC | sv284@duke.edu | 862-287-2223 | LinkedIn | GitHub

## Executive Summary

---

I am an undergraduate student at Duke University majoring in Physics and Math. I aspire to complete a PhD in physics to pursue a career in Quantum Information Science. Currently, I am interested in quantum materials, algorithms, and information theory, particularly where they intersect with condensed matter theory. I have developed my experience with QIS through immersing myself in research, building my theoretical foundations, and sharpening my programming skills. I hope to complement my theoretical interests with hands-on experimental experience along the way.

## Education

---

**Duke University**, BS in Physics and Math

Aug 2024 – May 2027

**GPA:** 3.9/4.0

**Courses:** Graduate Quantum Information Theory, Quantum Mechanics I, Abstract Algebra

**Skills:** Python, Qiskit, PyTorch, Numpy, Scipy, Matplotlib, Pandas, Mathematica, Bash, Slurm, LAMMPS

**Languages:** English, Tamil, Spanish (conversational)

## Research Experience

---

**Undergraduate Researcher**, Duke University

August 2025 – Present

Advisor: Dr. Natalie Klco

- Executed parallelized Mathematica computations to investigate application of majorization techniques in detecting entanglement in scalar fields. (manuscript in-progress)
- Formulated mathematical proofs to rigorously validate numerical findings.
- Numerically computed scaling of entanglement between disjoint regions of scalar field in continuum limit.
- Evaluated custom methods against standard benchmarks to ensure research accuracy and reproducibility.

**Undergraduate Researcher**, Duke University

October 2025 – Present

Advisor: Dr. Travis Nicholson

- Engineered high-fidelity simulation of Gross-Pitaevskii Equation (non-linear PDE) w/ split-step Fourier method to investigate novel properties of Bose-Einstein condensate of indium (manuscript in-progress).
- Accelerated simulation performance by migrating legacy codebase to object-oriented, GPU-accelerated PyTorch framework (yielding a 13.5x speed-up), while maintaining documentation and version control.
- Assemble, calibrate, and test custom electronics (e.g, digital loop filter) for use in atomic physics experiments.

**Research Assistant**, Duke University

June 2026 – Present

Advisor: Dr. Gaurav Arya

- Implemented data-driven model to map orientational interfacial free energy landscape of soft matter systems.
- Parallelized MD simulations across several physical configurations via LAMMPS on HPC Slurm cluster.
- Calculated Potentials of Mean Force (PMF) with Umbrella Sampling to characterize system thermodynamics.

**Research Intern**, Argonne National Laboratory, SULI

May 2025 – Dec 2025

Advisor: Dr. Zain Saleem, Dr. Ji Liu

- Benchmarked quantum error detection protocol under biased noise models to enhance quantum compilers. (<https://arxiv.org/abs/2509.15505>)
- Conducted in-depth literature review, experimentation, and data analysis using Pandas.
- Motivated investigation of use of deep learning models for hardware noise characterization.

**Research Intern**, North Carolina State University

July 2023 – Aug 2023

Advisor: Dr. Lex Kemper

- Optimized quantum gates by scaling microwave pulses on superconducting qubits w/ Qiskit.
- Achieved 60% speed-up in operations, resulting in longer simulation of a quantum system.

## Publications

---

**QuantEM: The quantum error management compiler.**  
arXiv:2509.15505 [quant-ph]

Sep 2025

## Work & Leadership Experience

---

**President**, Duke Quantum Information Society Oct 2024 – Present

- Organizing inaugural quantum hackathon, **DuQuantum**, securing over **\$30,000** in sponsorship funding.
- Promote accessibility to QIS and foster collaborative learning environment by coordinating regular talks with industry and academic speakers, hosting reading groups, teaching student-led QIS course.

**Freelance Tutor** Feb 2025 - Present

- Tutor elementary through high school students.
- Subjects range from middle school math to multivariable calculus to introductory calculus-based physics.

**Art of Problem Solving Morrisville**, Teaching Assistant Oct 2023 - June 2024

- Guided students through notoriously challenging math problems.
- Substitute teacher for Prealgebra and Geometry.
- Assisted 1st Grade Math class: taught section of weekly lesson and guided through group activities.

**Middle School Science Fair Club**, Student Coach Oct 2020 - June 2024

- Mentored middle school students in designing science projects for science fair competitions.

## Projects

---

**YQuantum Hackathon**, Graph Coloring with Neutral Atom Quantum Computers April 2025

- Implemented an adiabatic quantum algorithm for a graph-coloring problem.
- Translated optimization problem into quantum Hamiltonian.
- Solutions provide insight into diversification of insurer portfolios based on wildfire risk.

**MIT Hackathon - iQuHack**, MaxCut with Variational Quantum Circuits January 2025

- Investigated the application of various variational ansatz circuits in solving MaxCut (with a constraint) on IonQ's quantum computers.
- Implemented adjusted objective Hamiltonian to isolate optimal solution.

## Presentations

---

**Automatic Quantum Error Detection with Pauli Checks** Aug 2025  
Argonne National Laboratory (SULI)

**Divergence of Assisted Entanglement in the Continuum Limit of Scalar Fields** April 2026  
Duke University Undergraduate Research Symposium

## Selected Coursework, Certifications, & Summer Schools

---

- **Coursework:** Real Analysis, Probability, Differential Equations, Graduate Solid State Physics, Stochastic Calculus\*, General Relativity\*
- **Certifications:** IBM Quantum Excellence: 2022 Fall Challenge, 2023 Spring Challenge, 2024 Global Summer School, Basics of Quantum Information

- **Summer Schools:**

Machine Learning 4 Fusion Summer School (William & Mary University) June 2026  
Intensive study of machine learning model architectures and their applications in nuclear fusion